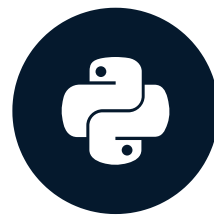


Memory and checkpointing in LangGraph

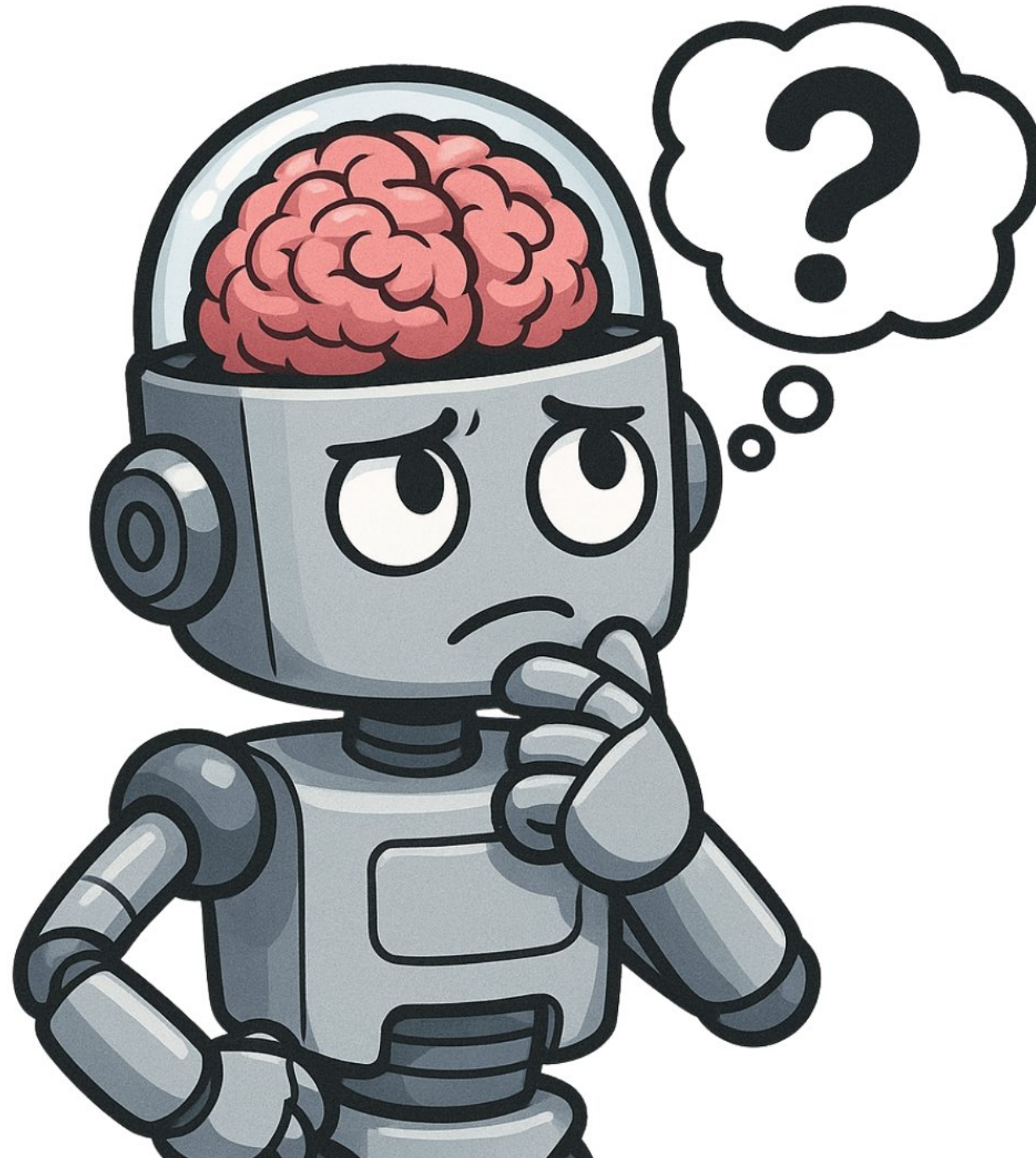
TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH

Apoorva Joshi

Senior AI/ML Developer Advocate,
MongoDB



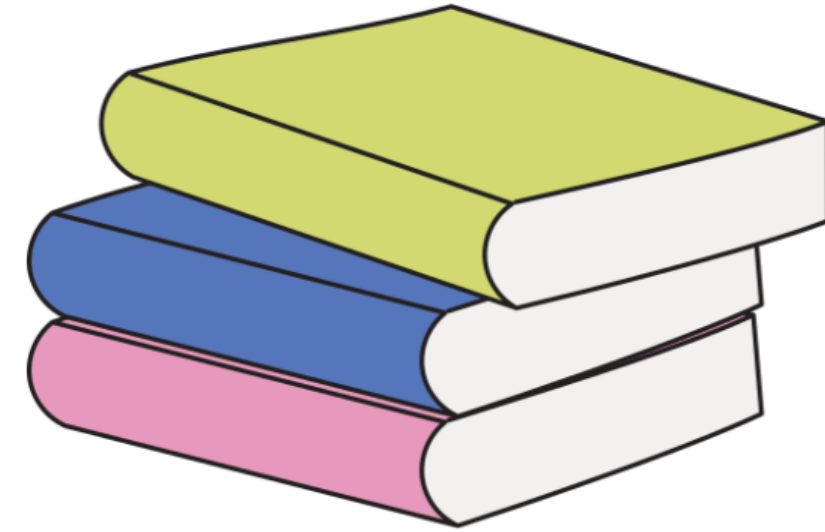
Never forget memory!





Short-term memory

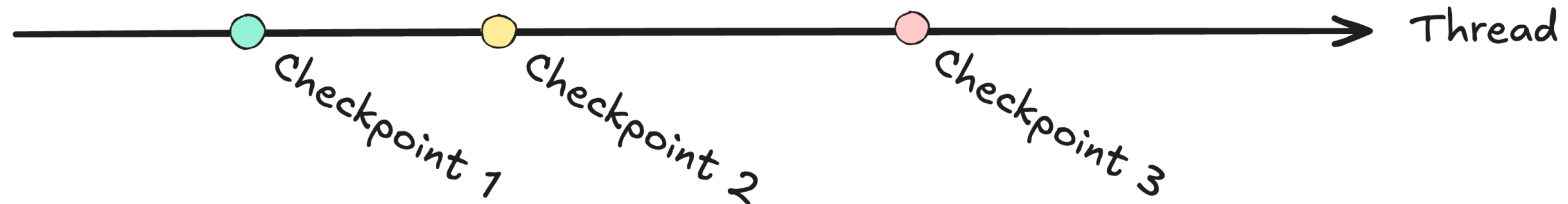
- Holds context during the conversation
- Usually doesn't persist beyond the session



Long-term memory

- Stored knowledge
- Can be recalled

Memory in LangChain



- **Checkpoints:** captures agent state at a given time
 - Typically *in-memory*
- **Thread:** the agent's conversation/workflow messages

Remembering memory



- `MongoDBSaver` → store checkpoints across runs
- Can resume from checkpoints
- **NOT** long-term memory!

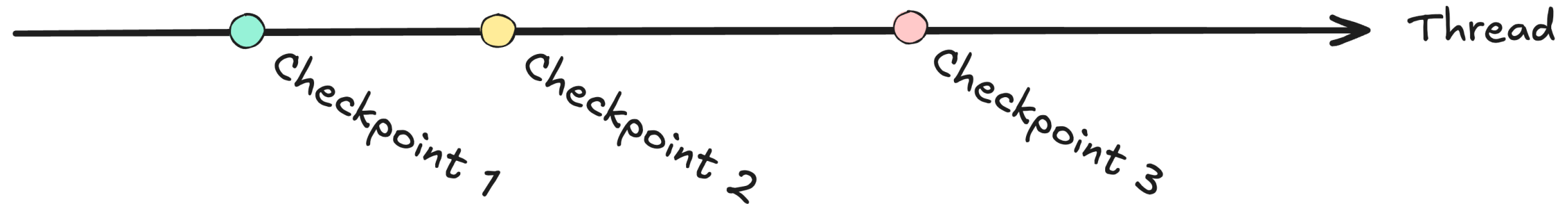
Remembering memory



- `MongoDBSaver` → store checkpoints across runs
- Can resume from checkpoints
- **NOT** long-term memory!
- `MongoDBStore` → long-term memory

¹ <https://www.mongodb.com/company/blog/product-release-announcements/powering-long-term-memory-for-agents-langgraph>

Summary



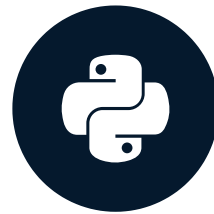
- **Short-term memory:** doesn't usually persist
- LangChain approach → *checkpoints* and *threads*
- `MongoDBSaver` persists checkpoints

Let's practice!

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Adding memory to the agent

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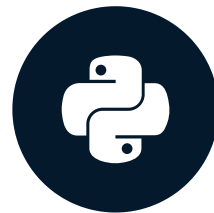
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Let's practice!

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Congratulations!

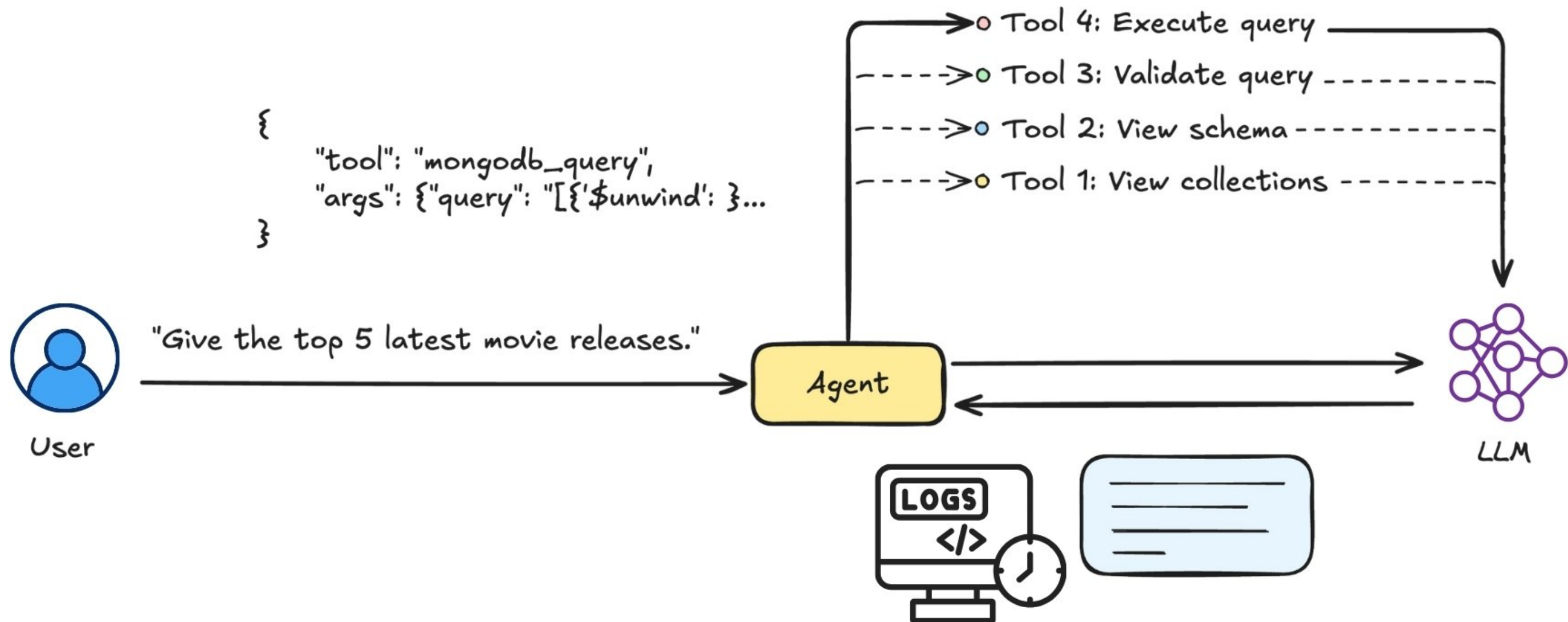
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The finished product!



Your agent's toolbox

```
from langchain_mongodb.agent_toolkit.toolkit import MongoDBDatabaseToolkit

# Initialize the MongoDB database toolkit
toolkit = MongoDBDatabaseToolkit(db=db, llm=llm)

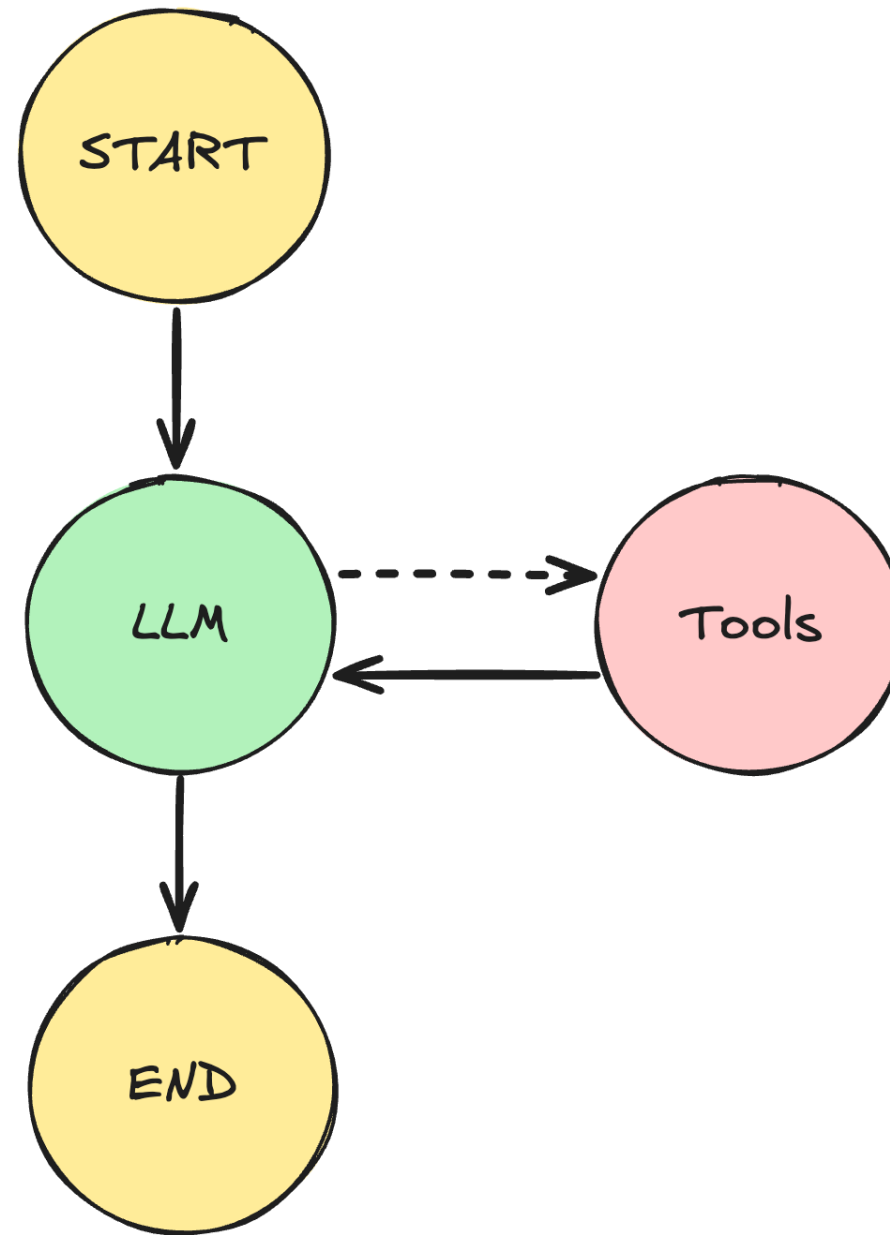
# Extract the tools from the toolkit
tools = toolkit.get_tools()
```

Prompt engineering

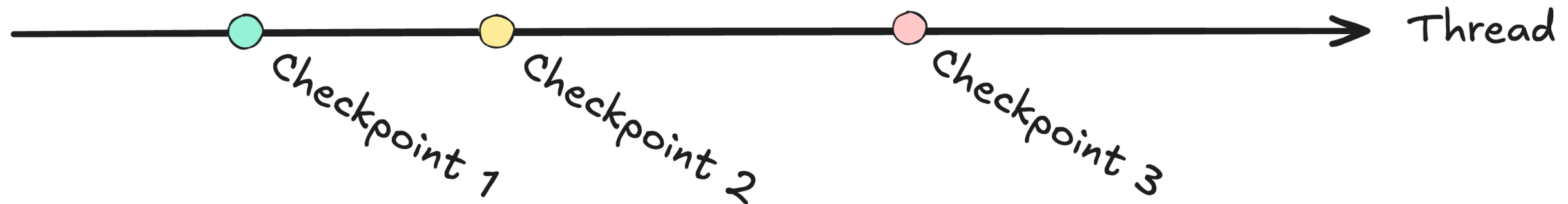
```
from langchain_mongodb.agent_toolkit import MONGODB_AGENT_SYSTEM_PROMPT

# Create a templated prompt for the LLM
prompt = ChatPromptTemplate.from_messages(
    [
        ("system", MONGODB_AGENT_SYSTEM_PROMPT),
        ("system", "Do not re-run tools unless absolutely necessary. If you are not
            able to get enough information using the tools, reply with
            I DON'T KNOW. You have access to the following tools: {tool_names}."),
        MessagesPlaceholder(variable_name="messages"),
    ]
)
```

Agent orchestration



Memory



What next?

More on MongoDB and Text-to-Query Agents

- Blog: [Text-to-Query Agents Using MongoDB](#)
- Blog: [Powering Long-Term Memory for Agents With LangGraph and MongoDB](#)
- [MongoDB's Gen AI examples GitHub repo](#)
- [MongoDB's AI Learning Hub](#)

More on LangChain and LangGraph

- DataCamp Course: [Developing LLM Applications with LangChain](#)
- DataCamp Course: [Designing Agentic Systems with LangGraph](#)

Congratulations!

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH